

Investigating the role of perceived stress on bacterial flora activity and salivary cortisol secretion: a possible mechanism underlying susceptibil...



This study examined the impact of academic stress on salivary cortisol concentrations and lactic acid bacteria activity. Whole, unstimulated saliva samples and faecal samples were collected from 23 healthy undergraduate students (23.0 ± 6.8 years; range 18-44) over two 1-week periods: during the beginning of semester (low-stress baseline condition) and during the first week of exams (high-stress condition). Students also completed a series of questionnaires measuring perceived levels of stress, gastrointestinal symptoms, and nutritional intake. Significant findings indicated that faecal lactic acid bacterial levels were lower during the high-stress condition. Paralleling this, students rated perceived levels of stress as being greater during the exam period compared to the baseline condition. The findings from this study have provided further insight into the link between stress and gastrointestinal flora activity in humans.